

ALFAAJ64232

## 5'-O-(4,4'-Dimethoxytrityl)-2'-fluoro-N2-isobutyryl-2'-deoxyguanosine

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

产品说明: 5'-O-(4,4'-二甲氧基三苯甲基)-2'-氟-N2-异丁酰基-2'-脱氧鸟苷  
Product Description: 5'-O-(4,4'-Dimethoxytrityl)-2'-fluoro-N2-isobutyryl-2'-deoxyguanosine

Cat No. : J64232  
CAS No 144089-96-3  
Molecular Formula C35 H36 F N5 O7

Supplier Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
Office Fax: +44 (0) 1524 850608

Emergency Telephone Number For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

E-mail address begel.sdsdesk@thermofisher.com

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Solid

**Appearance**  
White - Light yellow

**Odor**  
No information available

**Emergency Overview**  
Harmful if swallowed. Causes skin irritation. Causes serious eye irritation. May cause respiratory irritation.

#### Classification of the substance or mixture

|                                                    |            |
|----------------------------------------------------|------------|
| Acute Oral Toxicity                                | Category 4 |
| Skin Corrosion/Irritation                          | Category 2 |
| Serious Eye Damage/Eye Irritation                  | Category 2 |
| Specific target organ toxicity - (single exposure) | Category 3 |

#### Label Elements



Signal Word

Warning

## 5'-O-(4,4'-Dimethoxytrityl)-2'-fluoro-N2-isobutyryl-2'-deoxyguanosine

**Hazard Statements**

H302 - Harmful if swallowed  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H335 - May cause respiratory irritation

**Precautionary Statements****Prevention**

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P330 - Rinse mouth  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

Harmful if swallowed. Causes skin irritation. Causes serious eye irritation. May cause respiratory irritation.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants.

This product does not contain any known or suspected endocrine disruptors. Toxic to terrestrial vertebrates.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                                                             | CAS No      | Weight % |
|-----------------------------------------------------------------------|-------------|----------|
| 5'-O-(4,4'-Dimethoxytrityl)-2'-fluoro-N2-isobutyryl-2'-deoxyguanosine | 144089-96-3 | <=100    |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

None reasonably foreseeable.

**Self-Protection of the First Aider**

No special precautions required.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Avoid dust formation. Do not get in eyes, on skin, or on clothing.

**Storage**

Keep refrigerated.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

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## Exposure Controls

## Engineering Measures

None under normal use conditions. Ensure that eyewash stations and safety showers are close to the workstation location. .

## Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particle filter

**Small scale/Laboratory use** Maintain adequate ventilation

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                   |                          |                                          |
|-----------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                 | White - Light yellow     |                                          |
| <b>Physical State</b>             | Solid                    |                                          |
| <b>Odor</b>                       | No information available |                                          |
| <b>Odor Threshold</b>             | No data available        |                                          |
| <b>pH</b>                         | Not applicable           |                                          |
| <b>Melting Point/Range</b>        | No data available        |                                          |
| <b>Softening Point</b>            | No data available        |                                          |
| <b>Boiling Point/Range</b>        | No information available |                                          |
| <b>Flash Point</b>                | No information available | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>           | Not applicable           | Solid                                    |
| <b>Flammability (solid,gas)</b>   | No information available |                                          |
| <b>Explosion Limits</b>           | No data available        |                                          |
| <b>Vapor Pressure</b>             | No data available        |                                          |
| <b>Vapor Density</b>              | Not applicable           | Solid                                    |
| <b>Specific Gravity / Density</b> | No data available        |                                          |
| <b>Bulk Density</b>               | No data available        |                                          |
| <b>Water Solubility</b>           | No information available |                                          |

## 5'-O-(4,4'-Dimethoxytrityl)-2'-fluoro-N2-isobutryl-2'-deoxyguanosine

|                                         |                          |       |
|-----------------------------------------|--------------------------|-------|
| Solubility in other solvents            | No information available |       |
| Partition Coefficient (n-octanol/water) |                          |       |
| Autoignition Temperature                | No data available        |       |
| Decomposition Temperature               | No data available        |       |
| Viscosity                               | Not applicable           | Solid |
| Explosive Properties                    | No information available |       |
| Oxidizing Properties                    | No information available |       |

|                   |                 |
|-------------------|-----------------|
| Molecular Formula | C35 H36 F N5 O7 |
| Molecular Weight  | 657.69          |

## SECTION 10. STABILITY AND REACTIVITY

|                          |                                 |
|--------------------------|---------------------------------|
| Stability                | Stable under normal conditions. |
| Hazardous Reactions      | None under normal processing.   |
| Hazardous Polymerization | No information available.       |
| Conditions to Avoid      | None known.                     |
| Materials to avoid       | No information available.       |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Hydrogen fluoride.

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

|                                        |                                                           |
|----------------------------------------|-----------------------------------------------------------|
| (a) acute toxicity;                    |                                                           |
| (b) skin corrosion/irritation;         | Category 2                                                |
| (c) serious eye damage/irritation;     | Category 2                                                |
| (d) respiratory or skin sensitization; |                                                           |
| Respiratory                            | No data available                                         |
| Skin                                   | No data available                                         |
| (e) germ cell mutagenicity;            | No data available                                         |
| (f) carcinogenicity;                   | No data available                                         |
|                                        | There are no known carcinogenic chemicals in this product |
| (g) reproductive toxicity;             | No data available                                         |
| (h) STOT-single exposure;              | Category 3                                                |
| Results / Target organs                | Respiratory system                                        |
| (i) STOT-repeated exposure;            | No data available                                         |
| Target Organs                          | None known.                                               |

## 5'-O-(4,4'-Dimethoxytrityl)-2'-fluoro-N2-isobutryl-2'-deoxyguanosine

(j) aspiration hazard; Not applicable  
Solid

Symptoms / effects, both acute and delayed No information available

**SECTION 12. ECOLOGICAL INFORMATION**

**Ecotoxicity effects** Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

**Persistence and Degradability** No information available

**Bioaccumulative Potential** No information available

**Mobility in soil** No information available

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS**

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

**SECTION 14. TRANSPORT INFORMATION**

**Road and Rail Transport** Not Regulated

**IMDG/IMO** Not regulated

**IATA** Not regulated

**Special Precautions for User** No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Taiwan (TCSI), Japan (ISHL), New Zealand (NZIoC), Japan (ISHL).

## National Regulations

## SECTION 16. OTHER INFORMATION

**Prepared By** Health, Safety and Environmental Department  
**Revision Date** 17-May-2024  
**Revision Summary** New emergency telephone response service provider.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Legend**

|                                                                                                                              |                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>CAS</b> - Chemical Abstracts Service                                                                                      | <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                   |
| <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances | <b>DSL/NDSL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                  |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                                                    | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                       |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                                             | <b>AICS</b> - Australian Inventory of Chemical Substances                                         |
| <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                 |
| <b>WEL</b> - Workplace Exposure Limit                                                                                        | <b>TWA</b> - Time Weighted Average                                                                |
| <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     | <b>IARC</b> - International Agency for Research on Cancer                                         |
| <b>DNEL</b> - Derived No Effect Level                                                                                        | <b>PNEC</b> - Predicted No Effect Concentration                                                   |
| <b>RPE</b> - Respiratory Protective Equipment                                                                                | <b>LD50</b> - Lethal Dose 50%                                                                     |
| <b>LC50</b> - Lethal Concentration 50%                                                                                       | <b>EC50</b> - Effective Concentration 50%                                                         |
| <b>NOEC</b> - No Observed Effect Concentration                                                                               | <b>POW</b> - Partition coefficient Octanol:Water                                                  |
| <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              | <b>vPvB</b> - very Persistent, very Bioaccumulative                                               |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association                         | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code |
| <b>ADR</b> - European Agreement Concerning the International Carriage of Dangerous Goods by Road                             | <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships               |
| <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         | <b>ATE</b> - Acute Toxicity Estimate                                                              |
| <b>BCF</b> - Bioconcentration factor                                                                                         | <b>VOC</b> - (Volatile Organic Compound)                                                          |

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

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**End of Safety Data Sheet**